



HET-CD-003

(Power Processing Unit)

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Section 1 Component Scope

In order to provide the proper power to each of the hall effect thruster (HET) subsystems there needs to be a circuit to do so. This will be known as the power processing unit. This is commonly found on spacecraft to perform the task described previously.

1.1 Overview

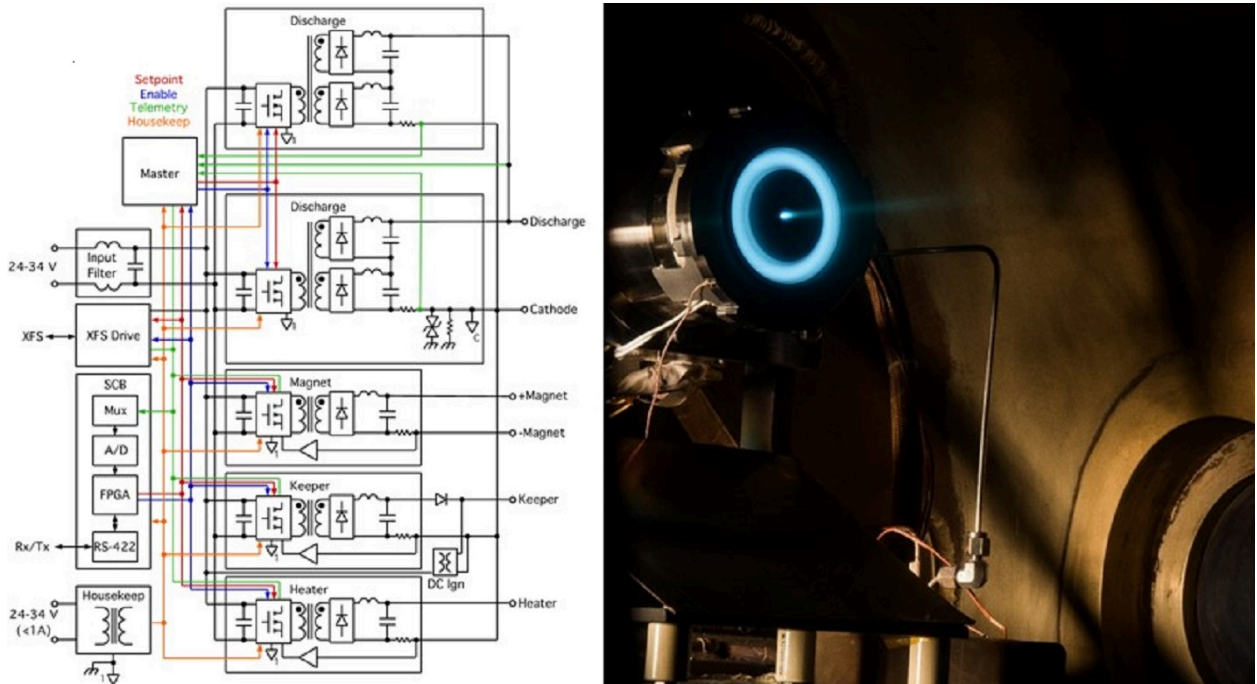


Figure 1: Example Diagram of PPU from NASA Low-Power HET

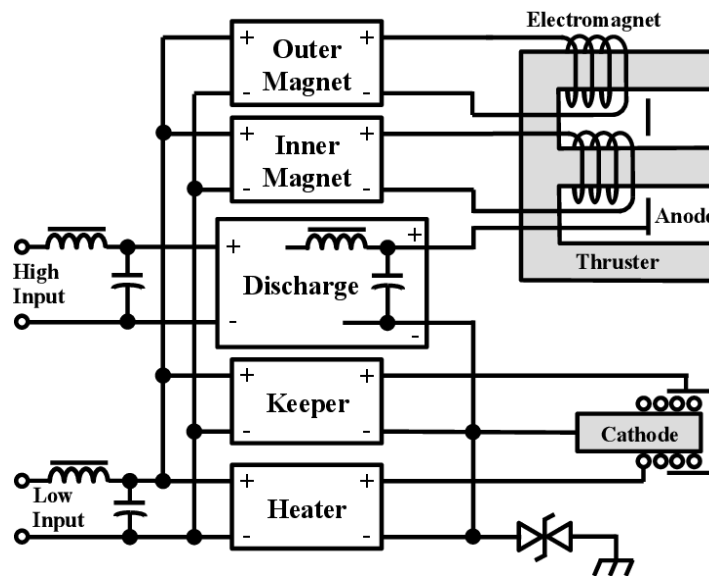


Figure 2: Example of Simplified Block Diagram for PPU of High-Power HET

1.2 Role in Assembly

The Power Processing Unit (PPU) distributes the power supply's output to subsystems within the HET. Each subsystem requires a different voltage and power input; the PPU provides this.

1.3 Design Objectives

The key objective is to provide reliable and safe power to the subsystems. It must be able to withstand all thermal loads and stress/vibrations.

Section 2 Requirements and Constraints

This section outlines the requirements and constraints necessary for the PPU.

2.1 Performance Requirements

The peak input power will be around 1300 Watts at 48 Volts. With a target efficiency of 80%. The power supply will be operated for no more than 10 minutes during that time it needs to be able to actively change its output to each subsystem in order to maintain optimal operating conditions. And it also needs to be able to take user inputs and then change its outputs based on those inputs.

2.1.1 Primary discharge circuit

The input voltage will be 48V and the outputs will be 0.8-2A at 300-400V. The plan is to overshoot these requirements by a smidge so as to give us a margin of error that we can use to tune the power supply to our needs. The primary discharge current and voltage needs to be adjustable for testing and experimentation of the thruster.

2.1.2 Coil Power supply

The coils need to be supplied with 0.5-1 amp of current; the resistance of one coil is estimated at .42 ohms (will probably change as the current simulation models progress.) This means the power supply will need to supply a max of .42 V. PPU will have a 48V input therefore the supply will need to step this voltage down and have an adjustable output.

2.1.3 Keeper supply

The keeper supply will need to deliver a pulse of 1 Amp at around 600 Volts(The voltage is dependent on the gas flow rate). It will then have to switch to a steady state mode where it delivers tens of volts. The exact steady state current and voltage; and the initial current and voltage pulse will be determined through testing.

2.2 Physical Constraints

It must be able to be attached to the test stand, and have enough power connections to receive and distribute power. It also needs to be enclosed so as to enable safe handling of the board

2.3 Manufacturing Constraints

The PPU will be assembled in-house. Components will be SMD and through hole, designed and selected with integration in mind via cross-team collaboration.

2.4 Environmental Constraints

The PPU will be housed outside of the vacuum chamber so it will need to be able to operate at STP.

2.5 Budgetary Constraints

There is no budgetary constraint as of now.

Section 3 System Design

The preliminary planning will be done using napkin calculations to get the general idea for potential resistor, inductor, capacitor, and transformer values based on the required channel power output. It will then be simulated in LTspice and the final board will be designed using Altium.

3.1 Geometry and Dimensions

3.1.1 Primary Discharge Circuit

The primary discharge current will be supplied by a multiphase boost converter. A multiphase boost converter was chosen for its capability to supply the required output voltage and current at a high efficiency. It will consist of five phases; each phase will lead the previous phase by $1/5$ th of a period. The calculated control frequency is 20Khz. The output capacitor is 13 uF. The mosfets are [R6020PNJs](#). They have a max Vds of 600V, Vgs of 30V, and a pulsed current of 80A.

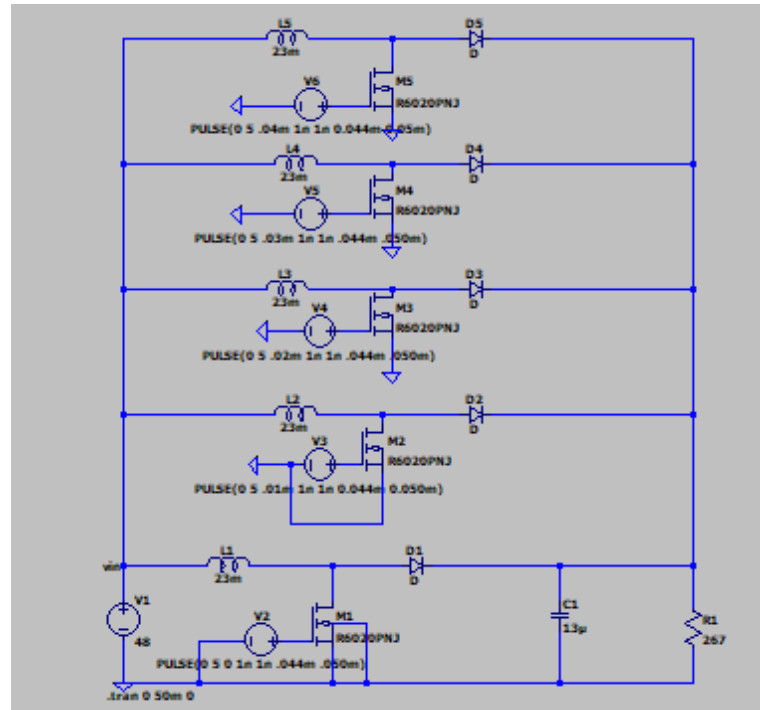


Figure 3: Primary Discharge Circuit LTSpice

Above is the LTSpice circuit diagram for the primary discharge circuit. R1 is the simulated load. The load is expected to vary, but we don't really know by how much at this time. The output voltage can be controlled by changing the duty cycle of the signals supplied to the mosfets. The control signals will be generated by the PPU's control unit. The control unit will have to account for changes in the plasma's resistance and adjust the boost converters duty cycle accordingly.

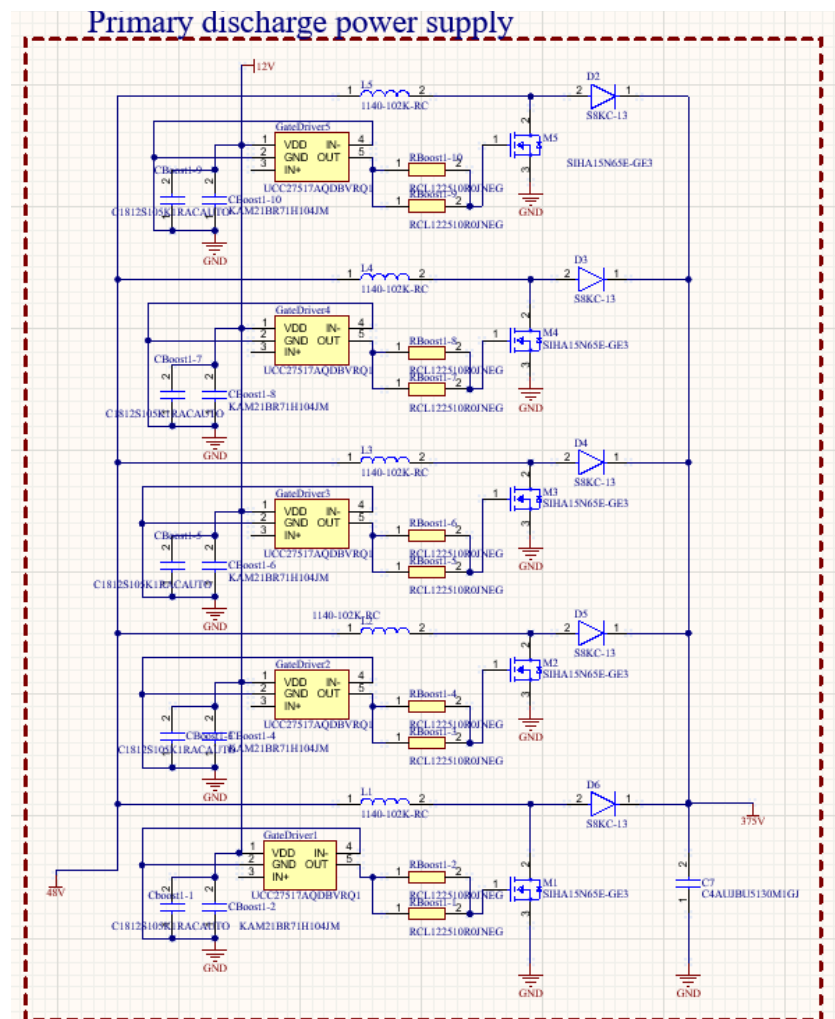


Figure 4: Primary Discharge Circuit (Altium)

Above is the circuit diagram of the primary discharge circuit in Altium. There are some differences in terms of the components chosen. For the current design the [R6020PNJ](#)s were swapped out for the SIHA15N65E-GE3. They have similar specifications; it's just that the SIHA15N65E-GE3 has better availability at the moment. Another key difference is the addition of a gate driver IC. Since the controllers logic output is 3.3 volts it won't be able to generate a great enough GS voltage to push the mosfets Vgs to its max on its own, that's where the gate driver comes in. It will take the controller's input signal and amplify it to 12 volts which will be enough to push our mosfets Vgs.

3.1.2 Magnetic Coil Circuit

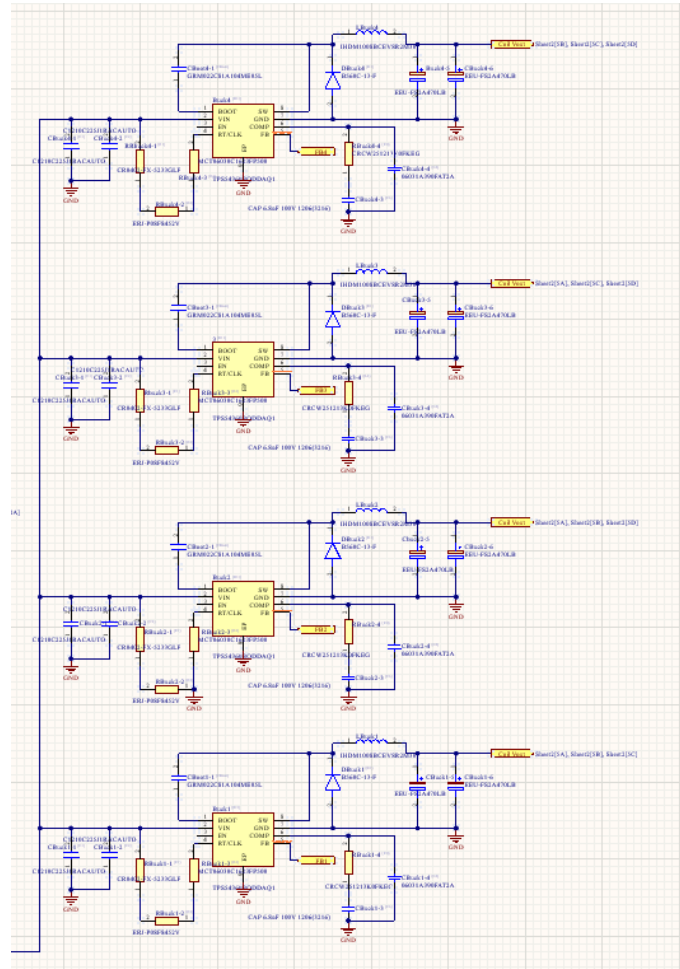


Figure 4: Magnetic Coil Power Circuit

Above is the full circuit diagram for the circuit that will be powering the magnetic coils. We are using the TPS54360BQDDAQ1 from Texas instruments which is a buck converter module. The entire coil power supply is a collection of four buck converters each feeding one of the four magnetic coils. Each buck converter can be controlled individually via their FB pin. When the FB pin is brought to .8 Volts it will hold the current output voltage steady. If it is below .8 volts then it will try and bring the output voltage up, if it is above it will try to bring the output voltage down .

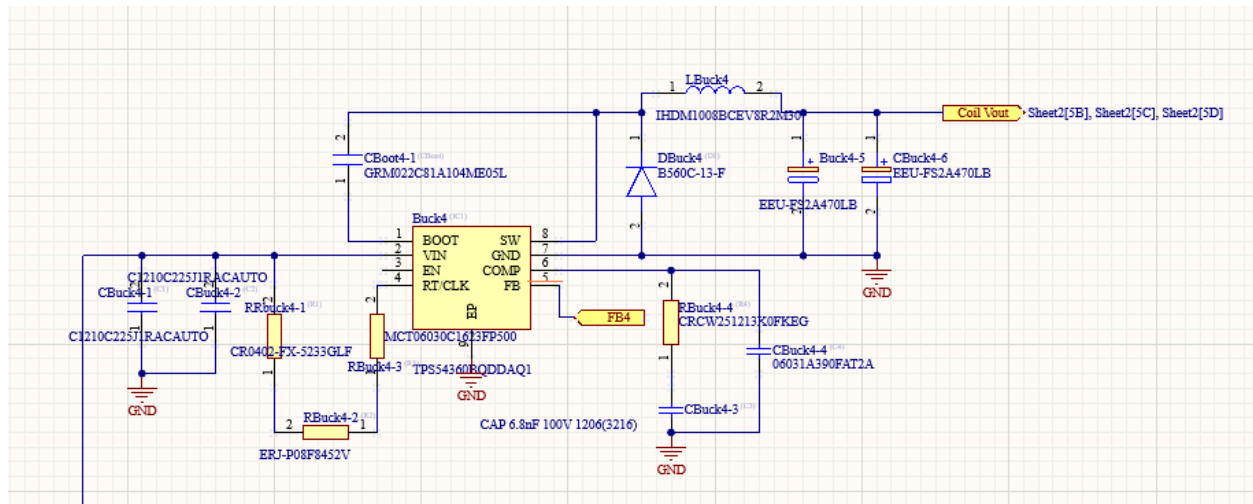


Figure 5: Magnetic Coil Power Circuit

Above is the individual circuit diagram for the buck converter.

3.1.3 Keeper Circuit

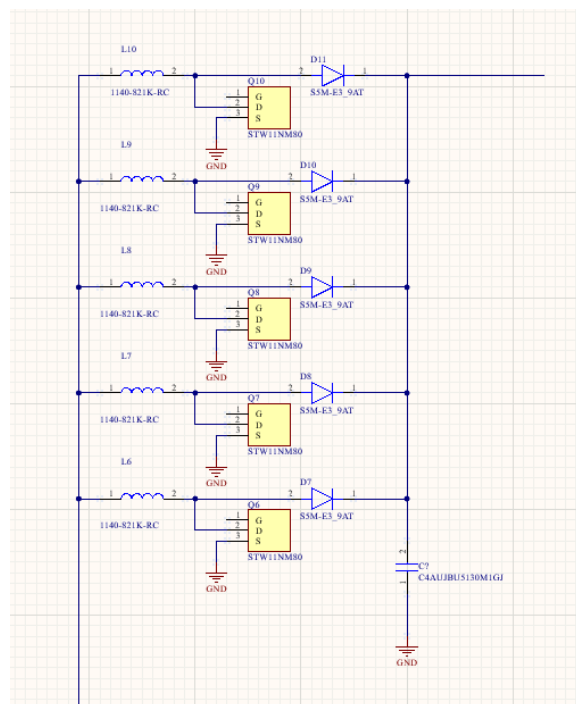


Figure 6: Keeper Power Circuit

The above keeper circuit is almost identical in operation to the primary discharge circuit. It takes five square wave signals as control inputs whose duty cycles are increased or decreased depending on the desired voltage.

3.2 Material Selection

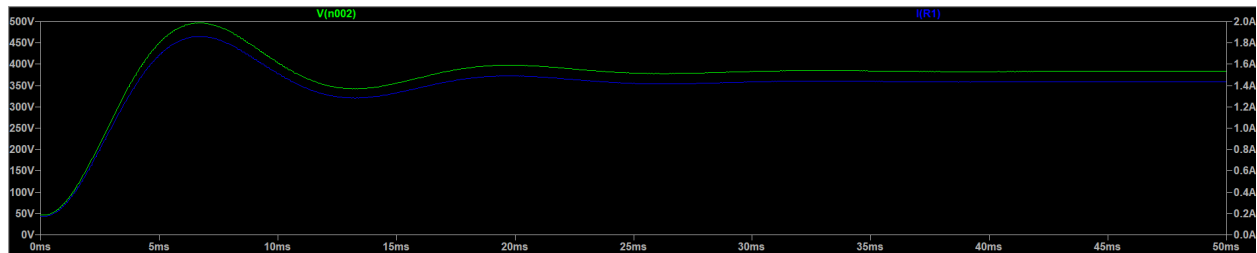
The PPU will use primarily off the shelf components that are listed in the bill of materials. Electrolytic capacitors and other components or materials that could be hazardous in a vacuum should be avoided. Even though the PPU will most likely not be inside the vacuum chamber it may need to be and therefore few changes to get it vacuum ready would be ideal.

3.3 Interface Definition

The PPU will need to be screwed into its enclosure therefore it will need screw holes to interface with M5 screws. The PPU will need to interface with an external computer to be programmed and or controlled, ideally via USB-C. The PPU will need power connectors to connect to its 48V power supply. The PPU will need to interface with the 50 pin Sub-D connector, that is the vacuum chambers pass through the connection(This could potentially be an issue as the keeper requires 600 volts but the sub d is only rated for 500 volts).

3.4 Analysis and Validation

The discharge module of the power supply was simulated in LTSpice. The simulation met the given specifications with an efficiency of 95%. A full write up of the simulation can be found in the electrical teams documentation.



Section 4 Manufacturing Plan

An electrical schematic will be sent to a PCB manufacturer with all necessary dimensions, and component placements.

4.1 Fabrication Method

We will have the boards printed by a third party vendor and then we will assemble the components onto the board by means of a heat gun and soldering iron.

4.2 Facility Capabilities

JLPCB will print the boards and we can solder them wherever soldering irons and the accompanying tools can be found. Coren personally should have all of the equipment needed to assemble the board.

Section 5 Testing and Verification

Once the PCB is received by the club from the manufacturer, tests will be conducted using a multimeter and the ESP32 to determine functional code and proper power output loads from the subsystem channels.

5.1 Pre-Test Requirements

Proper PPE must be worn when handling the PPU as physical contact with one of the high voltage terminals may result in death. No part of a person's body should be uncovered if handling the PPU while power is being supplied. The PPU should be fully disconnected from all power sources before being moved or handled by a person. When powered no one should be within the vicinity of the PPU and the area around it should be free of any potential hazards.

5.2 Verification Tests

Ensure that all the power load output channels are functioning as intended (all subsystems need specific voltage and current values tested using an oscilloscope or other device). Test MCU and FPGA code to see if commands are properly sent and interpreted (start up sequence, open and closing of solenoids).

Section 6 Safety and Risk Mitigation

6.1 Safety & Risk Mitigation

There is a high risk of electrocution and death. Always ensure that proper PPE is worn.

A circuit to discharge any high voltage capacitors when the ppu is not will be implemented.

Section 7 Cost Analysis

7.1 Bill of Materials

Item	Quantity	Total Price(unit price)	Link
CR0402-FX-5233GLF	1	\$0.10	https://www.mouser.com/ProductDetail/Bourns/CR0402-FX-5233GLF?qs=sGAEpiMZZMtubZb

			dhIBIK1I%2F7FNVFNK5I sWdnnZ1Uw%3D
ERJ-P08F8452V	1	\$0.29	https://www.mouser.com/ProductDetail/Panasonic/ERJ-P08F8452V?qs=sGAEpiMZZMtubZbdhIBIK7FbVYKWV6WVUxJQ35ttf0%3D
MCT06030C1623FP500	1	\$0.16	http://mouser.com/ProductDetail/Vishay-Beyschlag/MCT06030C1623FP500?qs=sGAEpiMZZMtubZbdhIBIMi2voYIIRG89DAjm1Idels%3D
C1210C225J1RACAUT0	2	\$1.04(\$0.52)	https://www.mouser.com/ProductDetail/KEMET/C1210C225J1RACAUT0?qs=s6nfgz3miVwiaTK1FL9dbA%3D%3D
GRM022C81A104ME05L	1	\$0.14	https://www.mouser.com/ProductDetail/Murata-Electronics/GRM022C81A104ME05L?qs=sGAEpiMZZMsh%252B1woXyUXjwVqLNpRsvnQXE8lta5ZOF8%3D
CRCW251213K0FKEG	1	\$0.33	https://www.mouser.com/ProductDetail/Vishay/CRCW251213K0FKEG?qs=sGAEpiMZZMtubZbdhIBIIF05JjrlZnUx73CQhI0Gjl%3D
C1608C0G2A682J080AC	1	\$0.20	https://www.mouser.com/ProductDetail/TDK/C1608C0G2A682J080AC?qs=sGAEpiMZZMsh%252B1woXyUXj6cbJqqnAu83ua5i45P4HIU%3D
12061C682KAT2A	1	\$0.30	https://www.mouser.com/ProductDetail/KYOCERA-AVX/12061C682KA

Section 8 References

Document No.	Document Title	Date	Author

<https://electricrocket.org/IEPC/IEPC-2009-117.pdf> - This is a great reference for what a PPU is and how power is distributed to the other subsystems. - Matthew Fan

<https://sqpcb.com/design-custom-circuit-boards/> - This should be a good introductory resource on how to design a PCB. - Max Day

Section 9 Appendices

Appendix A –